Improving Melanoma Classification Across Diverse Skin Tones Through Photometric Image Augmentation

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ABSTRACT

Automated melanoma classifiers can match dermatologist-level accuracy on curated benchmarks, yet they frequently underperform on darker skin tones because public dermatology datasets lack skin-tone diversity. This work investigates whether skin-tone-aware photometric image augmentation can improve classification robustness and fairness without designing a new architecture. An EfficientNet-B0 classifier, pretrained on ImageNet and fine-tuned on HAM10000, is evaluated as a cross-dataset generalization test on the Diverse Dermatology Images (DDI) set, which provides Fitzpatrick skin-tone labels. Two models are compared: a baseline trained with preprocessing only, and an augmented model trained with brightness, contrast, saturation, hue, flip, and rotation jitter. Grad-CAM overlays provide explainability. On DDI, augmentation improved malignant detection (F1 0.046 → 0.198, ROC-AUC 0.554 → 0.615) and narrowed the maximum–minimum per-group accuracy spread (0.077 → 0.049), but it did not reduce the Light-versus-Dark accuracy gap. The results suggest photometric augmentation is a weak proxy for genuinely diverse training data and that fairness gains most likely require representative images rather than synthetic color perturbation.

Index Terms—melanoma classification, skin-tone fairness, data augmentation, EfficientNet, Grad-CAM, computational imaging, medical image analysis.

I. INTRODUCTION

A. Problem and Motivation

Melanoma is one of the most severe forms of skin cancer, and early diagnosis significantly improves patient outcomes [1]. Advances in computational imaging and deep learning have enabled automated classification of skin lesions from clinical and dermoscopic images, and convolutional neural networks (CNNs) have become the dominant approach, demonstrating diagnostic performance comparable to dermatologists in some studies [1].

Despite these advances, existing skin-lesion classifiers often fail to generalize across diverse patient populations. A major limitation is the lack of skin-tone diversity in publicly available dermatology datasets [2]. Darker skin tones are frequently underrepresented, making it difficult to evaluate whether models perform consistently across all populations [2]. As a result, many systems exhibit reduced accuracy on skin-of-color (SOC) populations [2], [3]. Because patients with skin of color are already diagnosed at later stages, these disparities carry real clinical consequences, motivating the need for more equitable dermatology AI.

Recent work shows that state-of-the-art dermatology AI models perform worse on diverse clinical image sets containing darker skin tones, and that fine-tuning with more representative data reduces this gap [4]. This suggests improving dataset diversity may be as important as improving the model itself. Rather than developing a new classifier, this project investigates whether skin-tone-aware image augmentation — generating realistic variation in skin pigmentation and imaging conditions while preserving lesion characteristics — can improve melanoma classification across skin-tone subgroups.

B. Contributions

This work makes three contributions. First, it provides a reproducible cross-dataset pipeline that fine-tunes EfficientNet-B0 on HAM10000 and evaluates on the skin-tone-labeled DDI benchmark. Second, it offers a controlled comparison of a baseline versus a photometrically augmented model on both overall performance and per-skin-tone fairness. Third, it uses Grad-CAM–based explainability to verify that the model attends to lesions rather than surrounding skin or imaging artifacts.

II. RELATED WORK

Deep CNNs trained on dermoscopic datasets have achieved dermatologist-comparable accuracy for melanoma recognition [1], and the HAM10000 dataset [6] has become a standard training corpus of multi-source dermatoscopic images with biopsy-confirmed malignancies. Despite this progress, reviews of skin-lesion datasets show pervasive under-representation of darker skin tones, which limits fairness evaluation [2], and broader analyses of AI in dermatology for skin of color highlight both the promise and the persistent challenges of equitable performance [3]. The Diverse Dermatology Images (DDI) benchmark [4], [5] was curated specifically to expose these disparities; it is biopsy-confirmed and matched across Fitzpatrick skin types, and demonstrates that leading models — including one trained on HAM10000 — lose substantial ROC-AUC on Fitzpatrick V–VI skin, while fine-tuning on diverse data closes the gap [4]. To interpret model behavior, Grad-CAM produces class-discriminative localization maps from a trained classifier without extra labels, and is widely used to confirm that medical-imaging models focus on clinically relevant regions rather than artifacts [7].

III. METHODS

A. Datasets and Labels

HAM10000 is used for training/validation and DDI [5] as a held-out test set — a deliberate train/test domain shift (dermoscopic → clinical photography). Labels are binarized to Benign vs Malignant (HAM10000 `dx == mel` → Malignant; DDI `malignant` field). DDI Fitzpatrick codes are mapped to three

groups: Light (FST I–II), Medium (FST III–IV), and Dark (FST V–VI). Table I summarizes the splits.

TABLE I. Dataset Summary

Dataset	Total	Malignant	Benign	Skin tone
HAM10000 (train)	8012	884	7128	—
HAM10000 (val)	2003	229	1774	—
DDI (Light)	208	49	159	FST I–II
DDI (Medium)	241	74	167	FST III–IV
DDI (Dark)	207	48	159	FST V–VI

Images are converted to RGB, resized to 224×224, and normalized with ImageNet statistics (mean [0.485, 0.456, 0.406], std [0.229, 0.224, 0.225]). The HAM10000 train/validation split is a stratified 80/20 split performed by `lesion_id` so that no lesion appears in both splits (preventing leakage), with seed 42.

B. Model and Training

The classifier is EfficientNet-B0 [8], initialized with weights pretrained on the ImageNet dataset [9], which enables transfer learning by leveraging visual features such as edges, textures, shapes, and color patterns learned from millions of labeled natural images. The final classification layer was then replaced with a two-class head, Linear(1280 → 2), and fine-tuned for benign-versus-malignant skin lesion classification (Fig. 1). Only the head is new; the backbone is fine-tuned. Training uses class-weighted cross-entropy (inverse-frequency weights to counter benign-heavy imbalance), the Adam optimizer (learning rate 1×10^{-4}), and a `ReduceLROnPlateau` scheduler on validation F1. The best checkpoint is selected by validation F1, and training early-stops when validation F1 stops improving.

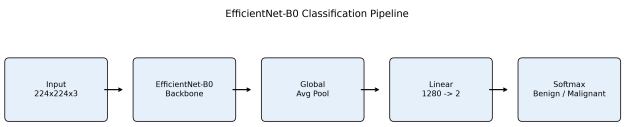


Fig. 1. End-to-end pipeline: image preprocessing, EfficientNet-B0 backbone, two-class head, and the Grad-CAM branch on the final convolutional block.

C. Skin-Tone-Aware Augmentation

Augmentation is applied to the training set only in the augmented experiment, before normalization, in random order: `RandomHorizontalFlip(0.5)`, `RandomRotation(±20°)`, and `ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2, hue=0.1)`. The color operations sample factors from uniform ranges and transform the image as follows, where `in`/`out` are pixel values, $\bar{g}$ is mean luminance, and `gray` is the grayscale image:

$$\text{brightness: } \text{out} = b \cdot \text{in}, \quad b \sim U(0.8, 1.2) \tag{1}$$

$$\text{contrast: } \text{out} = c\,(\text{in} - \bar{g}) + \bar{g}, \quad c \sim U(0.8, 1.2) \tag{2}$$

$$\text{saturation: } \text{out} = s\,\text{in} + (1 - s)\,\text{gray}, \quad s \sim U(0.8, 1.2) \tag{3}$$

$$\text{hue: shift } H \text{ in HSV by } h \sim U(-0.1, 0.1)\ (\approx \pm 36°) \tag{4}$$

Darkening arises specifically from a brightness factor $b < 1$ (and, secondarily, reduced contrast/saturation), approximating darker-skin appearance in the absence of skin-tone labels in HAM10000. Representative outputs are shown in Fig. 2.

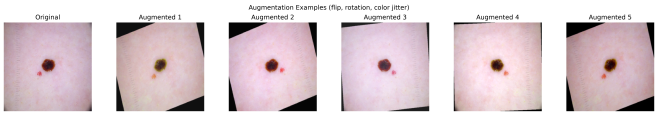


Fig. 2. One training lesion rendered under several augmentations (flip, rotation, color jitter).

D. Grad-CAM Explainability

Grad-CAM is computed on the last convolutional block (`model.features[-1]`). For a target class c with activation maps A^k (channel k), the gradient of the class score y^c is global-average-pooled to channel weights α_k^c, combined, and rectified:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{5}$$

$$L^c = \text{ReLU}\Big(\sum_k \alpha_k^c A^k \Big) \tag{6}$$

L^c is bilinearly upsampled to 224×224, min–max normalized to $[0, 1]$, and alpha-blended over the input as $(1 - \alpha)\,\text{image} + \alpha\,\text{jet}(L^c)$ with $\alpha = 0.4$.

IV. EXPERIMENTAL SETUP

Two experiments are run with identical evaluation: baseline (preprocessing-only training) and augmented (Section III-C). Both are trained on HAM10000 and evaluated on all 656 DDI images. Hyperparameters: batch size 32, Adam learning rate 1×10^{-4}, seed 42, up to 15 epochs with early stopping on validation F1. With batch size 32, one epoch is 251 training batches (8012 images) and 63 validation batches (2003 images). Metrics are Accuracy, Precision, Recall, F1, and ROC-AUC (positive class = Malignant), plus per-skin-tone accuracy and a Light–Dark fairness gap. Training was CPU-bound; the baseline early-stopped near epoch 6 (best validation F1 at epoch 3). All artifacts (checkpoints, metrics JSON, figures, tables) are regenerated deterministically from the pipeline.

V. RESULTS

A. Overall Performance

On the DDI test set, augmentation substantially improved malignant detection. The baseline is extremely conservative — perfect precision but near-zero recall — whereas the augmented model trades some precision for much higher recall, F1, and ROC-AUC (Table II). ROC curves and confusion matrices are shown in Fig. 4 and Fig. 5; training curves in Fig. 3.

TABLE II. Overall DDI Metrics (Baseline vs Augmented)

Metric	Baseline	Augmented
Accuracy	0.745	0.741
Precision	1.000	0.512
Recall	0.023	0.123
F1	0.046	0.198
ROC-AUC	0.554	0.615

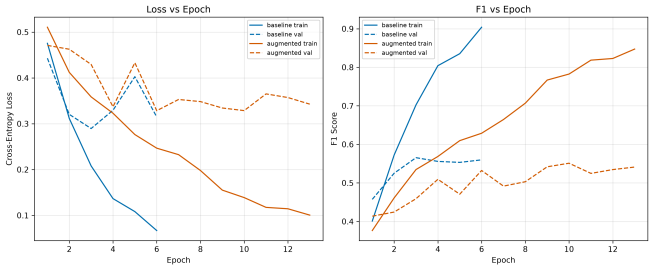


Fig. 3. Training/validation loss and F1 across epochs.

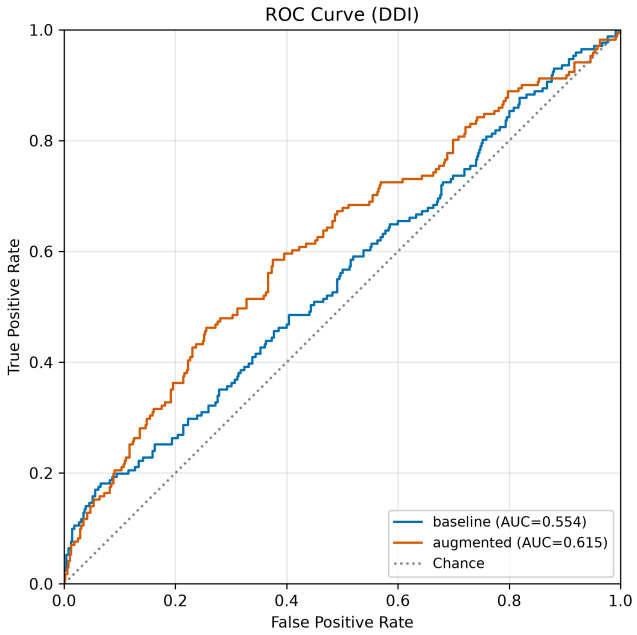


Fig. 4. ROC curves (DDI) for the baseline and augmented models.

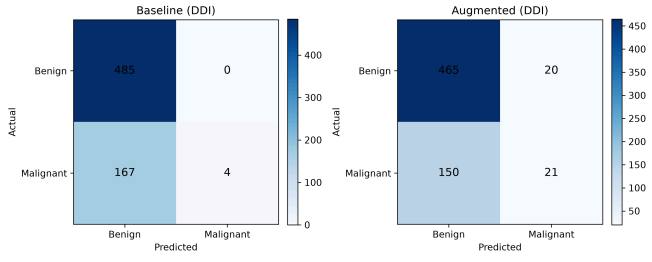


Fig. 5. Confusion matrices (DDI) for the baseline and augmented models.

B. Fairness Across Skin Tones

Per-group results are reported in Table III, and the fairness gaps in Table IV. Augmentation raised Light and Medium accuracy but lowered Dark accuracy (0.778 → 0.725). As a result, the overall max–min spread narrowed (0.077 → 0.049), but the primary Light–Dark gap did not improve and in fact became positive (−0.013 → +0.049). Per-group recall/F1 (Table III) did rise for every group, consistent with the overall sensitivity gain.

TABLE III. Per-Skin-Tone Metrics

Model	Skin tone	Count	Accuracy	Precision	Recall	F1
Baseline	Light	208	0.764	0.000	0.000	0.000
Baseline	Medium	241	0.701	1.000	0.027	0.053
Baseline	Dark	207	0.778	1.000	0.042	0.080
Augmented	Light	208	0.774	0.750	0.061	0.113
Augmented	Medium	241	0.726	0.722	0.176	0.283
Augmented	Dark	207	0.725	0.263	0.104	0.149

TABLE IV. Fairness Gap (Accuracy by Group)

Model	Light	Medium	Dark	Light−Dark	Max−Min
Baseline	0.764	0.701	0.778	−0.013	0.077
Augmented	0.774	0.726	0.725	+0.049	0.049

C. Explainability

Grad-CAM overlays (Fig. 6) across skin tones and both classes confirm the model generally concentrates activation on the lesion region rather than surrounding skin or imaging artifacts, supporting that predictions are lesion-driven.

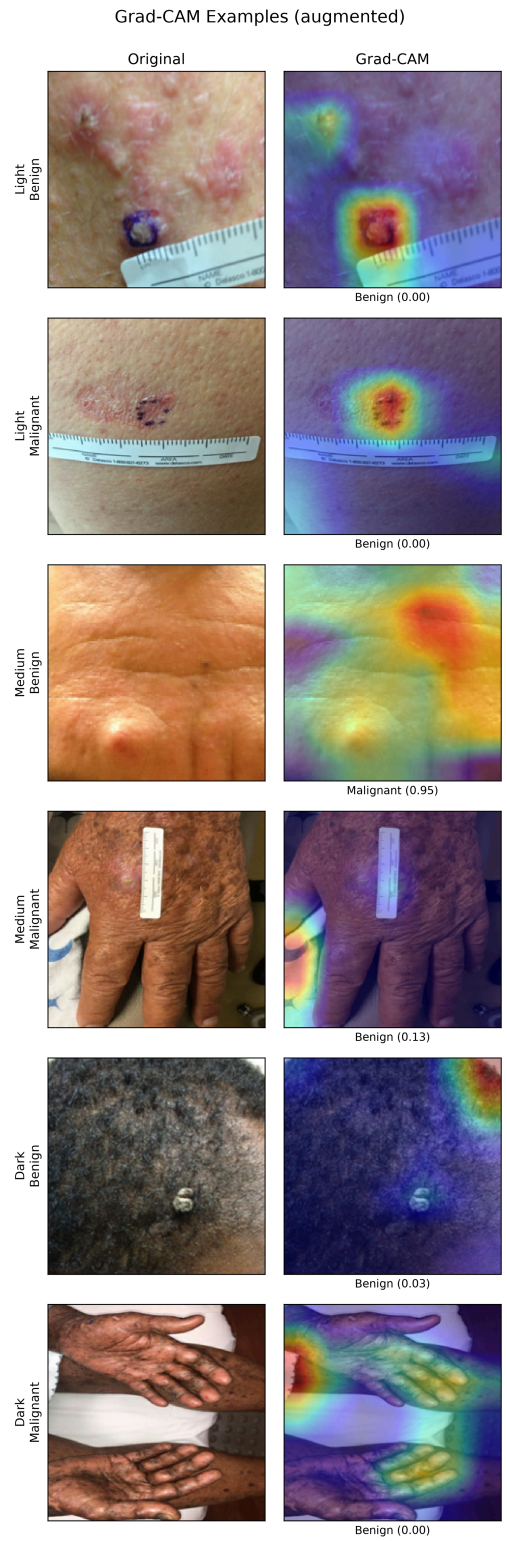


Fig. 6. Grad-CAM examples across skin tones (Light/Medium/Dark) and classes (Benign/Malignant).

VI. DISCUSSION

Augmentation clearly improved the model's ability to *detect* malignancy: recall rose from 0.023 to 0.123, F1 from 0.046 to 0.198, and ROC-AUC from 0.554 to 0.615. The baseline behaved as a near-constant "benign" predictor (perfect precision, negligible recall), which inflates accuracy on a benign-heavy test set while being clinically useless; augmentation produced a more balanced operating point.

The fairness picture is mixed. The max–min accuracy spread narrowed, but the targeted Light–Dark accuracy gap did not improve — Dark-group accuracy actually fell even though Dark-group recall and F1 increased. This apparent tension is a consequence of the benign-heavy class balance: accuracy and sensitivity move differently as the decision threshold shifts. The most plausible explanation is that photometric jitter changes pixel intensities but not lesion morphology or the characteristics of real skin-of-color clinical photography, so it is a weak proxy for genuine diversity.

This aligns with the DDI study [4], where the large fairness gains came specifically from fine-tuning on diverse DDI data (HAM10000 Light/Dark ROC-AUC rising to ≈0.77/0.78), not from color augmentation of a non-diverse corpus. Our augmented model never trains on DDI, so it should not be expected to reproduce those gains. Our baseline DDI ROC-AUC (0.554) is also lower than the study's HAM10000 baseline (0.67), attributable to a different architecture (EfficientNet-B0 vs Inception V3), a single seed, class-weighted loss versus balanced sampling, early stopping, and different preprocessing/decision thresholds.

An additional observation comes from the Grad-CAM visualizations. While several Light-skin examples show model attention concentrated around the lesion itself, some Medium- and Dark-skin examples exhibit activations that are partially displaced from the lesion region and extend into surrounding skin or background structures. This suggests that the model may not always localize diagnostically relevant features consistently across skin-tone groups. Although the number of Grad-CAM examples is too small to draw definitive conclusions, inconsistent localization could help explain why fairness improvements were limited despite improvements in overall sensitivity. If the model learns different visual cues for different skin-tone groups, performance disparities may persist even when overall classification metrics improve.

VII. LIMITATIONS

Several factors limit the scope and interpretation of these results.

The most fundamental limitation is that the training and test data come from different sources. HAM10000 consists primarily of dermoscopic images, whereas DDI contains clinical photographs. These datasets differ in acquisition conditions, image quality, patient populations, and label distributions. Consequently, the reported performance reflects both classifier quality and the difficulty of cross-dataset generalization.

A second limitation is the absence of skin-tone labels in HAM10000. Because skin-tone information is unavailable during training, the model cannot explicitly learn fairness-aware representations or be optimized for equitable performance across

demographic groups. Fairness can only be assessed afterward using DDI.

The augmentation strategy itself is also limited. Photometric jitter modifies pixel intensities but does not reproduce the anatomical, textural, and photographic characteristics of genuine skin-of-color clinical images. In particular, the ±0.2 brightness adjustment used in this project darkens images only modestly and may not adequately represent the appearance variability observed in real-world dermatology datasets. As a result, augmentation serves only as a weak proxy for true dataset diversity.

An additional limitation is revealed by the Grad-CAM visualizations. Although many examples show attention focused on lesion regions, several Medium- and Dark-skin examples exhibit activation patterns that are partially shifted away from the lesion and into surrounding skin or unrelated image regions. These observations suggest that lesion localization may be less consistent across skin-tone groups. Because accurate classification depends on identifying clinically relevant structures, inconsistent localization could contribute to the remaining fairness gap and may help explain why augmentation improved overall metrics while failing to consistently improve Dark-group performance. The current analysis includes only a small number of explainability examples, so this interpretation should be considered exploratory rather than conclusive.

Evaluation is further constrained by the relatively small size of DDI. Each skin-tone subgroup contains only approximately 200–240 images, which increases the variance of subgroup performance estimates and makes fairness metrics sensitive to small numbers of classification errors. In addition, all experiments were performed using a single random seed and CPU-limited training, preventing a more comprehensive statistical analysis.

Finally, Grad-CAM itself has limitations as an explainability tool. Its heatmaps indicate influential image regions but do not provide precise lesion boundaries and should not be interpreted as segmentation masks. The system is intended solely for research purposes and is not a medical device. Its predictions should not be used for clinical diagnosis or treatment decisions.

VIII. CONCLUSION AND FUTURE WORK

This project asked whether skin-tone-aware image augmentation can improve melanoma classification across diverse skin tones. Using EfficientNet-B0 trained on HAM10000 and evaluated on the skin-tone-labeled DDI benchmark, augmentation improved overall malignant detection (F1 and ROC-AUC) and narrowed the max–min per-group accuracy spread, but did not reduce the targeted Light–Dark gap. The hypothesis is therefore only partially supported: photometric augmentation helps robustness but is insufficient, on its own, to close skin-tone fairness gaps — consistent with prior findings that representative data is the key lever [4].

Future work could fine-tune on genuinely diverse data (as in [4]), apply stronger augmentation techniques, run multiple random seeds with confidence intervals, train on the full dataset with GPU acceleration, and investigate domain-adaptation approaches such as Deep CORAL [10] to reduce the distribution shift between HAM10000 and DDI. Fairness-aware optimization methods such as GroupDRO [11] may also be explored if skin-tone labels become available during training. Another promising direction is adopting datasets with pixel-level lesion masks, such as ISIC and PH2, to support segmentation-based approaches rather than Grad-CAM-only localization [12].

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